Detecting the Internal Condition of Tree Trunks using Acoustic Tomography Method

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***Abstract*—Urban trees are critical to environmental sustainability, providing oxygen, reducing greenhouse gases, and enhancing urban aesthetics. However, internal defects such as decay and cavities pose significant risks, including tree falls and structural failures. Current tree health assessment methods, such as visual inspection and Forest Health Monitoring (FHM), fail to examine internal trunk conditions, while existing advanced tools are often costly and inaccessible. This project addresses these limitations by designing and implementing an affordable, portable device utilizing the acoustic tomography method to detect internal defects in tree trunks. By measuring stress wave velocities and processing tomographic images, the device accurately visualizes internal wood density variations. The developed system achieves a wave velocity resolution of 41.05 m/s, a sampling rate of 28,600 Hz, and supports portability with an IP23-rated casing and a rechargeable battery lasting up to 3 hours. This innovation offers an accessible solution for arborists, researchers, and forestry professionals, enabling reliable non-invasive tree health assessments.**

Keywords—Acoustic Tomography, Tree Health, Non-Invasive, Wave Velocity, Portable Device

I. Introduction

Urban trees play a vital role in enhancing environmental quality and aesthetics by providing oxygen, reducing greenhouse gases, and serving as habitats for various species [1] [2]. However, internal defects such as decay and cavities pose serious risks, potentially leading to falls or breakages that cause damage and injury [3]. In environments with a high density of trees, it is crucial to have reliable methods for assessing tree health. Tree maintenance consultants, including tree researchers, arborists, and forestry experts, need to accurately understand both the external and internal conditions of trees.

Currently, tree researchers at Institut Teknologi Bandung (ITB) and similar environments rely on visual assessment and Forest Health Monitoring (FHM) to check tree health. These methods do not allow for direct observation of the internal condition of tree trunks. Existing instruments that can assess the internal health of trees are very expensive and not always accessible to researchers with limited budgets. Therefore, there is a need for an affordable, non-destructive tool that can accurately detect the internal condition of tree trunks based on density variations.

Design and implementation of the internal health of tree trunks detector product using the acoustic tomography method aim to provide researchers the capabilities to detect internal tree health decay by measuring its wood distribution densities. The device should be able to measure wave velocity with a resolution of 41,05 m/s, operate at a sampling rate of 35 µs (28.600 Hz), provide visualizations of wood decay through a mobile app readable at a distance of 41–46 cm, complete measurements in less than 1 hour per tree height point, and ensure portability with an IP23-rated casing and with a rechargeable battery lasting up to 3 hours.

II. Material and Methods

A. Characteristics of Acoustic Wave in Living Trees

The acoustic waves propagating within wood due to an impact can be referred to as stress waves. Stress waves travel very quickly in dense, solid wood. However, if there are cavities or cracks within the wood, the stress waves propagate more slowly. Stress waves do not pass through cavities or cracks directly but instead follow their contours, as illustrated in Fig 1. When the propagation path bends around the contours, the travel time of the stress waves increases. This phenomenon can indicate the internal health condition of the wood. High stress wave velocity within a tree suggests that the wood is solid and free from cavities. Conversely, low stress wave velocity indicates decayed or hollow wood.

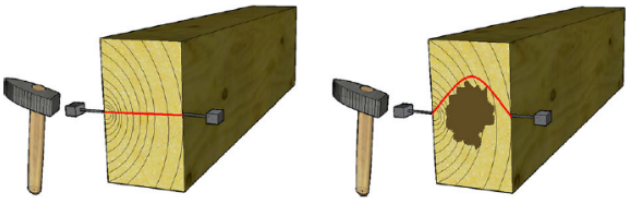


Fig. 1. Stress wave propagation in healthy wood (left) and decayed wood (right)

Based on their propagation direction relative to the tree's growth orientation, stress waves are classified into two types: longitudinal stress waves and transverse stress

waves, as shown in Fig 2. Longitudinal stress waves travel parallel to the tree's growth direction and exhibit high velocities, ranging from 3,500 m/s to 5,000 m/s. In contrast, transverse stress waves propagate perpendicular to the tree's growth direction and have lower velocities, ranging from 1,000 m/s to 1,500 m/s. Additionally, the propagation angle significantly affects stress wave velocity; larger angles result in reduced propagation speeds [4].



Fig. 2. Time-of-flight testing with (left) longitudinal excitation and (right) transverse excitation

B. Acoustic Tomography

One of the interpolation methods for tomography that has been developed is the ellipse method[5]. The area affected by a stress wave ray is an ellipse of different sizes. The size of this ellipse can be modeled by equation (5)

$$c = b/a \tag{5}$$

where b is the short-axis value, a is the long-axis value, and c is the control coefficient that satisfies equation (6)

$$c_{ij} = \{1 - \left[\frac{|j-i|}{\frac{N}{2}}\right],\ |j-i| < \frac{N}{2}\ 1 - [\frac{N-|j-i|}{\frac{N}{2}}],\ |j - \tag{6}$$

where i and j are sensor numbers and N is the total number of sensors.

Then, to determine the affected area inside the ellipse, we can use equation (7)

$$h(x,y) = \{1,\ \frac{D_{xb}^2}{a^2} + \frac{D_{ya}^2}{b^2} \leq 1\ 0,\ \frac{D_{xb}^2}{a^2} + \frac{D_{ya}^2}{b^2} > 1 \tag{7}$$

where D_{xb} is the distance of the point to the short axis of the ellipse, D_{ya} is the distance of the point to the long axis of the ellipse, and $h(x,y) = 1$ indicates that the point is inside the ellipse. The affected area representation generated by the ellipse method can be seen in Fig. 3.

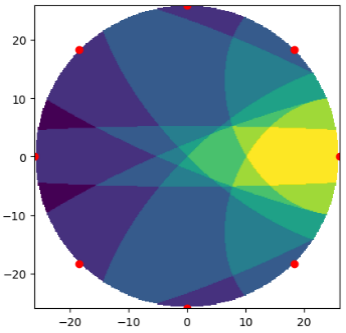


Fig. 3. Affected area using ellipse method

III. Design and Implementation

Acoustic tomography is the chosen method for detecting internal tree trunk health in this study. This section provides an overview of the principles, design, and methodology of acoustic tomography.

A. Principles

The principle of operation of acoustic tomography lies in the analysis of sound wave propagation through tree trunks. When acoustic waves encounter changes in density or structural anomalies within the wood, such as decay or cavities, they experience variations in velocity and amplitude. By analyzing these changes, it is possible to create internal images of the tree trunk. This method enables the detection of internal defects and anomalies without the need for invasive procedures[6].

B. Top-Level Architecture Design

Fig 4 shows the design architecture of the product. The product consists of 7 subsystems: Data Acquisition Subsystem, Control and Data Processing Subsystem, Communication Subsystem, Application Subsystem, Visualization Subsystem, Data Storage Subsystem, and Power Subsystem.

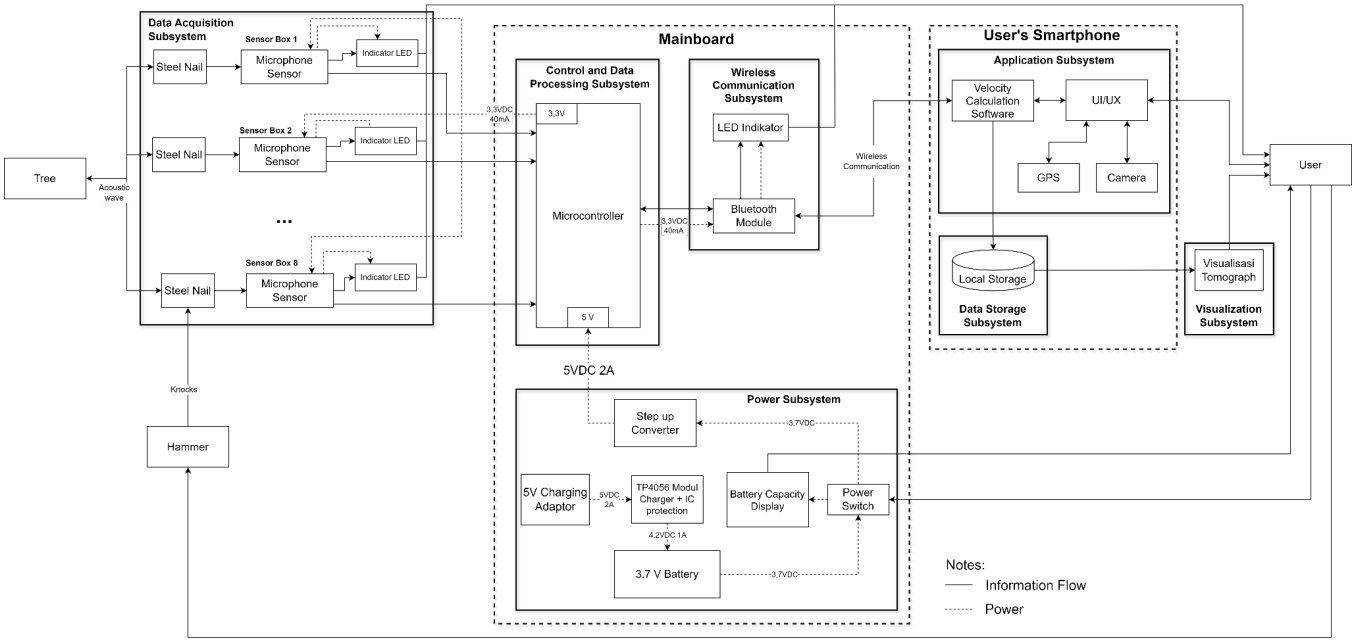


Fig. 4. *Architecture design*

In designing and implementing the product, the specifications must meet user needs. In the development of this product, we set the specifications based on user's needs as shown in Table 1.

TABLE I. PRODUCT DESIGN SPECIFICATIONS

Product Characteristics	Specification	Details
Measures tree trunk's wood density distribution	Wave Velocity Reading Resolution	41,05 m/s
Measures tree trunk's wood density distribution	Sampling Rate	35 µs or 28.600 Hz
Visualizes tree trunk decay	Measurement Display	Text and images displayed on the smartphone app is readable at a distance of 41–46 cm.
Short measurement time	Total Measurement Time	Less than 1 hour per one layer measurement
Portable and splash-resistant	Casing	IP23-rated casing and sensor nails are highly resistant to pressure, deformation, and corrosion.
Portable and splash-resistant	Battery Life	Rechargeable battery lasts for around 3 hours of operation in a single day.
Portable and splash-resistant	Product Weight and Dimensions	Fits in a luggage box, manageable weight for 1–2 adults, cables suited for trees with 120 diameter and up to 180 cm measurement height.
Stores measurement data	Memory Capacity	Stores data for at least 240 images for 20 working days.

C. Data Acquisition Subsystem

In the data acquisition subsystem, Time of Flight data is collected between each sensor. Acoustic waves read by the sensor cause an interrupt or sensor trigger. Then, the interrupt time is forwarded to the control subsystem, as seen in Fig 5.

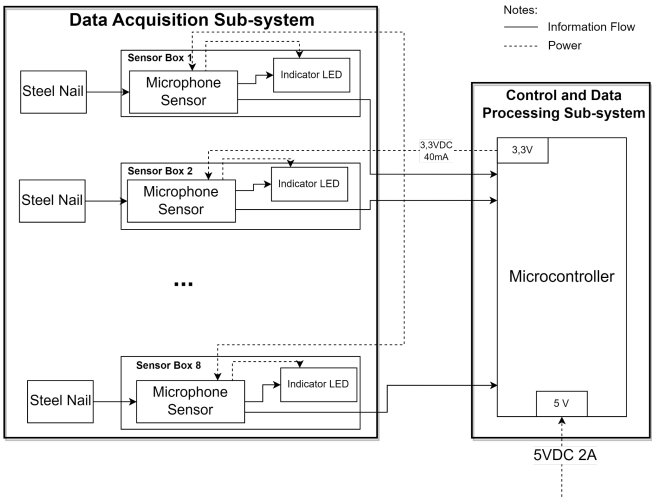


Fig. 5. Data acquisition subsystem decomposition diagram

D. Control and Data Processing Subsystem

In this subsystem, the microcontroller using Teensy 4.1 acts as the control center and connects the power lines from the power subsystem and each sensor. The measurement data read by the sensors is processed into an $N \times N$ matrix where N is the number of sensors used, as seen in Fig 6.

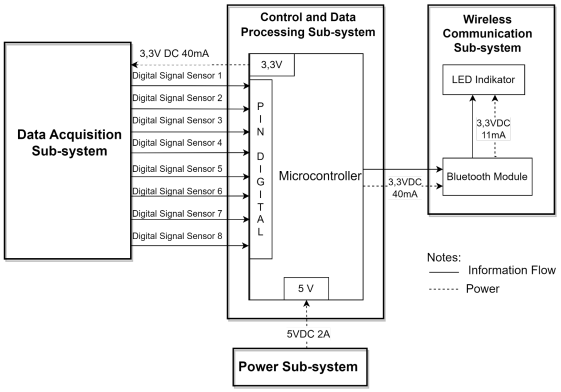


Fig. 6. Control and data processing subsystem decomposition diagram

E. Wireless Communication Subsystem

Then, the measurement data matrix enters the wireless communication subsystem. The data is then forwarded to the application subsystem using wireless communication. The module used in this process is a Bluetooth module, as seen in Fig. 7.

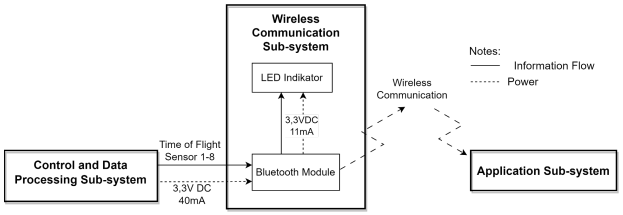


Fig 7. Wireless communication subsystem decomposition diagram

F. Application Subsystem

The Application Sub-System processes the time-of-flight data received from the Wireless Communication Sub-System into velocity data using the Velocity Calculation Software. This process is guided by user inputs provided through the UI/UX interface. Once the velocity data is calculated, all the measurement and tree data input will be saved to local storage by Data Storage Sub-System, as seen in Fig 8.

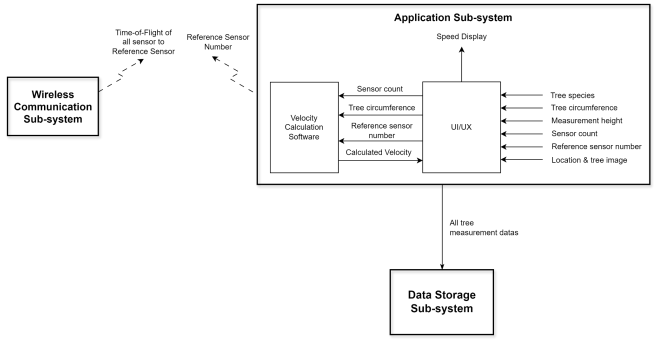


Fig. 8. *Applications subsystem decomposition diagram*

G. Data Storage Subsystem

Fig. 9. shows the data storage subsystem and visualization subsystem decomposition diagram.

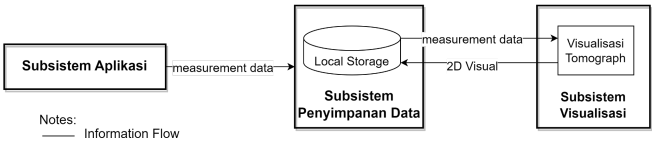


Fig. 9. Data storage subsystem and sisualization subsystem decomposition diagram

Variables of tree characteristics that have been entered by the user in the application as well as sensor measurement data are processed by the data storage subsystem. In this subsystem, a file is created containing tree species, tree size, measurement location, number of sensors, and wave velocity data.

H. Visualization Subsystem

In this subsystem, stress wave velocity is processed into a 2D visual. The method used in performing the reconstruction is tomography. The output of this subsystem is an image that can be displayed on the application screen.

I. Power Subsystem

The Power Subsystem ensures a stable and portable energy supply for the acoustic tomography device, as seen in Fig. 10. It consists of a 3.7V rechargeable lithium-ion battery managed by a TP4056 charging module with IC protection to prevent overcharging and short circuits. A step-up converter boosts the battery output to the required voltage for the system, while a power switch controls energy flow. This subsystem provides power to the sensors, microcontroller, and Bluetooth module, as seen in Fig 10. Additionally, a battery capacity display allows users to monitor charge levels in real-time. This design ensures up to 8 hours of operation, portability, and efficient power management.

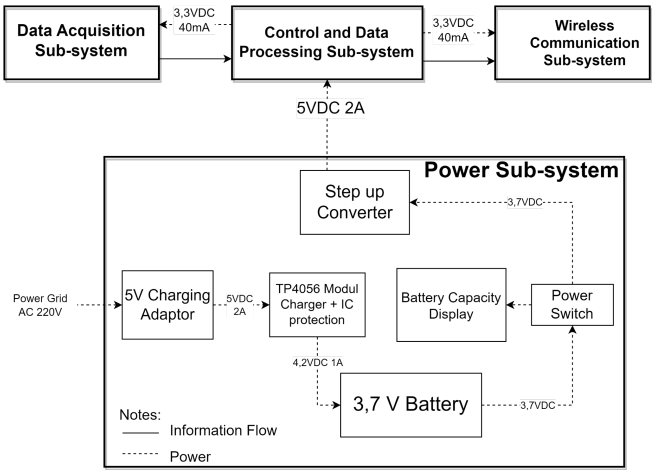


Fig. 10. Power subsystem decomposition diagram

IV. Results and Analysis

A. Product Integration

The integration of the Acoustic Tomography product is shown in Fig 11.

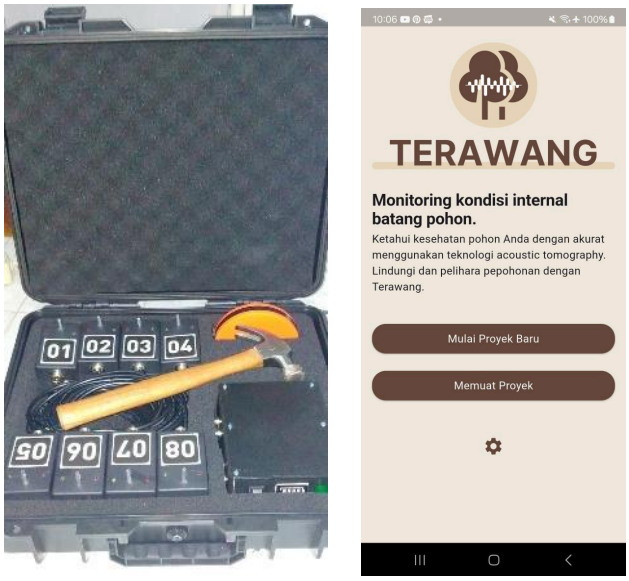


Fig. 11. (Left) Acoustic tomography equipment box (right) acoustic tomography mobile application

B. Wave Velocity Reading Resolution

The Wave Velocity Reading Resolution was tested by measuring acoustic wave propagation on wood with controlled defects. The verification was conducted as shown in Fig 12., where two sensors were placed 24 cm apart on the wood. Initially, a hole with a diameter of 2,4 cm (10% of the distance between the sensors) was drilled between the sensors, and the first sensor was tapped 10 times to record the time of flight (ToF) of the acoustic waves, which was averaged to determine the wave velocity. The average ToF was found to be 231,6 µs, resulting in a wave velocity of 1036,26 m/s. Subsequently, a second hole of the same diameter was drilled, making the total void area 20% of the distance between the sensors. After recording and averaging ToF data (excluding invalid readings), the ToF was 236,4 µs, yielding a wave velocity of 1015,22 m/s. The analysis showed a velocity difference of 21,04 m/s between the 10% and 20% void conditions. This value is significantly below the specified resolution threshold of 41,05 m/s.



Fig. 12. Setup using 10% hole (left) and 20% hole (right)

C. Sampling Rate

The Sampling Rate specification was tested by measuring the time taken for the device to read data from a sensor 20 times in succession. The test utilized a timer function to record the difference between the timestamps of the first and 20th readings, which was then divided by 20 to determine the average sampling time. The test program showed that the time for a single reading was t = (2301007-2301001) / 20 = 0,3 microsecond. With the device using 8 sensors in operation, the total time for one cycle of readings across all sensors is t = 8 × 0,3 = 2,4 microsecond, corresponding to a sampling frequency of f = 416,67kHz. This result exceeds the required specification of f > 28,6kHz or t < 35 microsecond. Fig 13. shows the sampling rate test results.

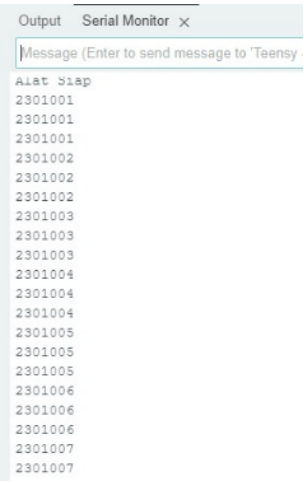


Fig. 13. Sampling rate test result using serial monitor

D. Measurement Display

Testing was conducted through a survey on the font and image display of the application screens using Google Forms with several users. Users were instructed to hold their phones at a distance of 41-46 cm from their eyes and view the screens of the mobile application. Based on the text and images seen on each screen, users filled out the Google Form with ratings ranging from 1 (unclear/not visible) to 5 (clearly visible).

Based on the survey results, the application has been able to provide users with a clear display of measurement results, especially due to the font size adjustment feature.

E. Total Measurement Time

The testing was done by demonstrating all stages of measurement, starting from opening the equipment case, obtaining measurement results and tomogram images for a single measurement height point, to packing all of the equipment back into the case. The duration of each measurement stage was timed using a stopwatch and compared to the target time, which was set at under 1 hour. Based on the testing, all measurement stages for a single tree height point were completed in 27 minutes and 40 seconds, meeting the total time specification of under 1 hour.

F. Casing

The casing was tested for water, solid object, and nail corrosion resistance, meeting IP23 standards. Placed under light rain for 8 minutes, the interior remained completely dry, demonstrating effective water protection. Tests with solid objects larger than 12.5 mm, such as a terminal block and screwdriver, confirmed the casing's resistance to external interference. The nails, hammered into a tree trunk 10 times and immersed in saline for 13 days, showed no

significant corrosion or deformation. These results confirm the casing and nails are durable and reliable for challenging field conditions. Fig. 14. and Fig. 15. shows the testing done for the verification.



Fig. 14. IP23 casing test result

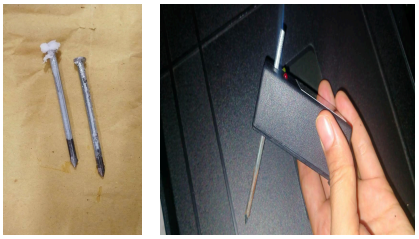


Fig. 15. Sensor nail test result

G. *Battery Life*

The Battery Life specification was tested using a 10.000mAh 3,7V Li-Po battery under a simulated load of 5,91W for 3 hours and 20 minutes. The test showed a consumed capacity of 3.238mAh and an average voltage of 3,915V, resulting in 12.671mWh of energy used—nearly double the device's requirement of 6.630mWh for 3 hours, as seen in Fig 16. After the test, the battery voltage recovered from 3,70V to 4,08V, confirming its health and efficiency. With an effective capacity of 29.600mWh, the battery can theoretically power the device for up to 13 hours, far exceeding the specification, ensuring reliable and long-lasting performance.

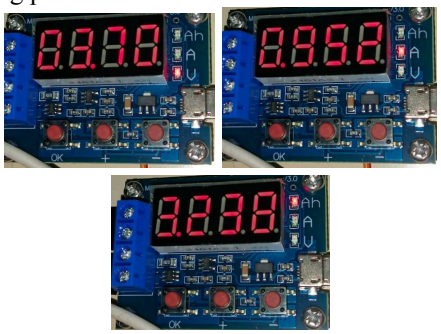


Fig. 16. Battery life test result

H. *Product Weight and Dimensions*

The product weight and dimensions specification was tested by evaluating the portability, weight, and cable suitability for measuring trees, as shown in Fig. 17. All components, including 8 sensor boxes, 8 cables (3 meters each), a hammer, a mainboard box, and a measuring tape, were placed in a wheeled luggage box measuring 42 x 36 x 12 cm with a total weight of 3,894 kg. The luggage box design includes a handle for easy carrying and can be lifted by 1-2 adults, ensuring portability. Cable suitability was tested on a tree with a maximum diameter of 100 cm and a measurement height of 180 cm, as no 120 cm diameter tree was available. The 3-meter cables were sufficient for this setup, meeting the intended requirements.



Fig. 17. Weight and dimension test result

I. *Memory Capacity*

The Memory Capacity specification was not tested, as the device relies on local storage provided by the connected smartphone. No standalone memory capacity evaluation was conducted. As this feature is considered an additional function rather than a core specification, its implementation was not prioritized during development. The reliance on smartphone storage ensures that sufficient memory is available for data recording and visualization without the need for separate memory modules within the device.

J. *Specification Analysis*

After completing the design and implementation process, we achieved a product with the characteristics and specifications detailed in Sections B to I. The next step involves tabulating and analyzing the obtained characteristics and specifications against the desired ones. The characteristics and specifications of the implemented "Detecting the Internal Health of Tree Trunks Using the Acoustic Tomography Method" are presented and detailed in Table II on the following page.

TABLE II. IMPLEMENTATION SPECIFICATIONS

Specification	Details	Details
Wave Velocity Reading Resolution	V	21,04 m/s
Sampling Rate	V	2,4 µs or 416,67 kHz
Measurement Display	V	Text and images on the smartphone app are clear and readable at a distance of 41–46 cm, with adjustable font size settings enhancing user visibility.
Total Measurement Time	V	27 minutes and 40 seconds per one layer measurement
Casing	V	IP23-rated casing and sensor nails are highly resistant to pressure, deformation, and corrosion.

Battery Life	V	The rechargeable battery can support the device for up to 13 hours of operation.
Product Weight and Dimensions	V	The device fits into a compact luggage box measuring 42 x 36 x 12 cm, with a total weight of 3,894 kg, making it manageable for 1–2 adults to carry. The included cables are suitable for measuring trees with a diameter of up to 100 cm and a height of 180 cm.
Memory Capacity	V	local storage provided by the connected smartphone

K. Test 1 (Using a cross-section of a dry tree)

For the first testing, a test medium in the form of a cross-section of a dry tree trunk is used, as shown in Fig. 18.



Fig. 18. A cross-section of dry tree trunk

In this test media, data was collected 5 times on each sensor. Table II shows the results of the average of acoustic waves velocity from 5 test data on each sensor.

TABLE II. THE AVERAGE OF ACOUSTIC WAVES VELOCITY ON TEST 1

Sensor	v1	v2	v3	v4	v5	v6	v7	v8
1	#	1101	1310	1228	1419	1425	1357	1231
2	1114	#	1113	1286	1287	1248	1019	1368
3	1537	1437	#	1295	1512	1450	1088	1481
4	1270	1313	1064	#	1028	1258	1371	1328
5	1636	1500	1589	1400	#	1268	1554	1583
6	1526	1347	1429	1395	1114	#	1129	1348
7	1460	1436	1340	1488	1416	1122	#	1074
8	1311	1415	1377	1363	1434	1300	1031	#

From the velocity data above, tomography processing is applied, resulting in the image shown in Fig. 19.

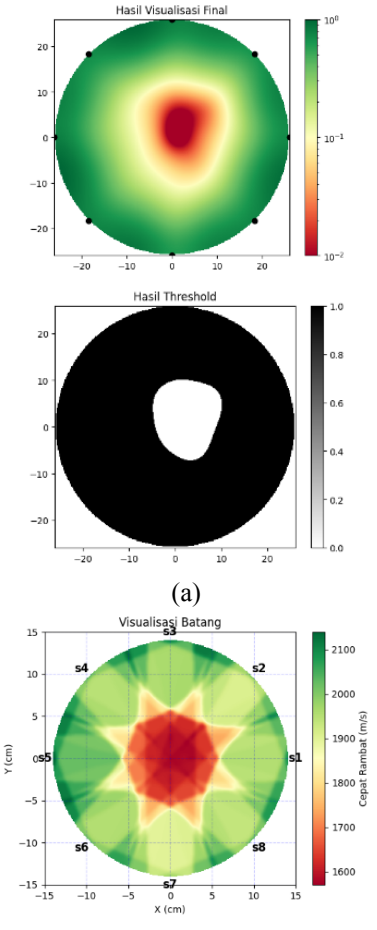


Fig. 19. (a) Tomography images on test 1 using thick rays method. (b) Tomography images on test 1 using the ellipse method.

The tomography image results successfully detect the presence of holes, as observed in the test media.

L. Test 2 (Using Living Tree)

The 2nd testing was conducted on a living tree, as shown in Fig. 20. The tree had a diameter of 39,8 cm and a circumference of 125 cm, measured at a height of 130 cm from the ground.



Fig. 20. Living tree on test 2

The average acoustic wave velocity results from Test 2 are shown in Table III.

TABLE III. THE AVERAGE OF ACOUSTIC WAVES VELOCITY ON TEST 2

Snsr	v1	v2	v3	v4	v5	v6	v7	v8
1	#	1046	1398	1233	1272	1364	1449	1178
2	1054	#	1179	1160	1444	1588	1692	1509
3	1725	1604	#	1318	1527	1722	1747	1861
4	1465	1410	1068	#	1032	1319	1066	1581
5	1561	1595	1398	1122	#	1103	1372	1501
6	1575	1721	1624	1440	1111	#	1095	1379
7	1549	1788	1742	1430	1351	1056	#	1158
8	1296	1586	1591	1381	1359	1190	1098	#

Tomogram results of Test 2 are shown in Fig. 21.

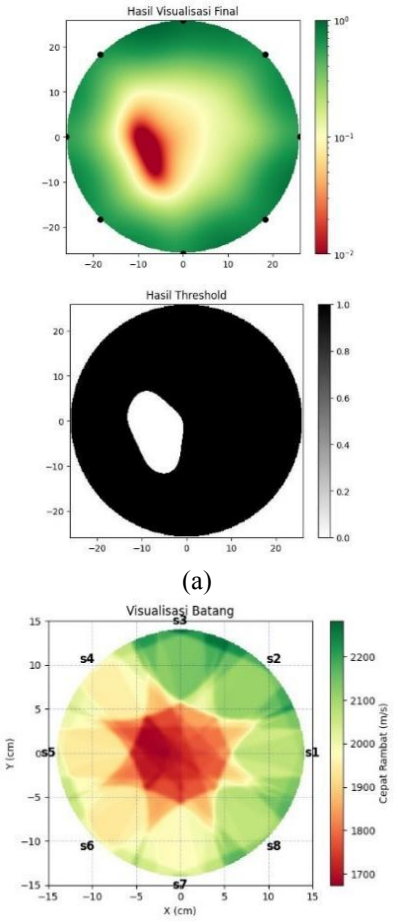


Fig. 21. (a) Tomography images on test 2 using thick rays method. (b) Tomography images on test 2 using ellipse method.

Based on the result, a decayed area was observed in the test tree trunk. The measurement data indicated 20% relative velocity decrease (RVD) in the area close to sensor 5 and sensor 6. Based on the RVD, it is estimated that there is 10%-20% decay in that area.

M. CONCLUSIONS

This research successfully developed a portable, affordable, and non-invasive acoustic tomography device to detect internal defects in tree trunks. Utilizing sound sensors (KY-037), the device demonstrated high accuracy in identifying cavities and low-density areas with a wave velocity resolution of 21.04 m/s, a sampling rate of 416.67 kHz, and a battery life of up to 13 hours. Field tests on both dry and living trees validated its effectiveness, particularly on hard and dry wood. However, performance was less reliable on wet or gummy wood, highlighting an area for potential improvement. With its compact and durable design, the device offers a practical and reliable solution for arborists and researchers to enhance tree health assessment and management in diverse environmental conditions.

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